

* Why Synchronization is needed? (Interview Question)
 ⇒ RACE Condition

A race condition occurs when multiple threads access shared memory/data simultaneously and produce incorrect results.

DeadLock ⇒ Deadlock occurs when two or more threads wait forever for each other's locks.

- * Circular Dependency
- * No thread can proceed
- * Program hangs or freezes

- * Write better code to avoid deadlock.
- * First let the 1st one finish locking & unlocking.
- * Then follow for the rest.

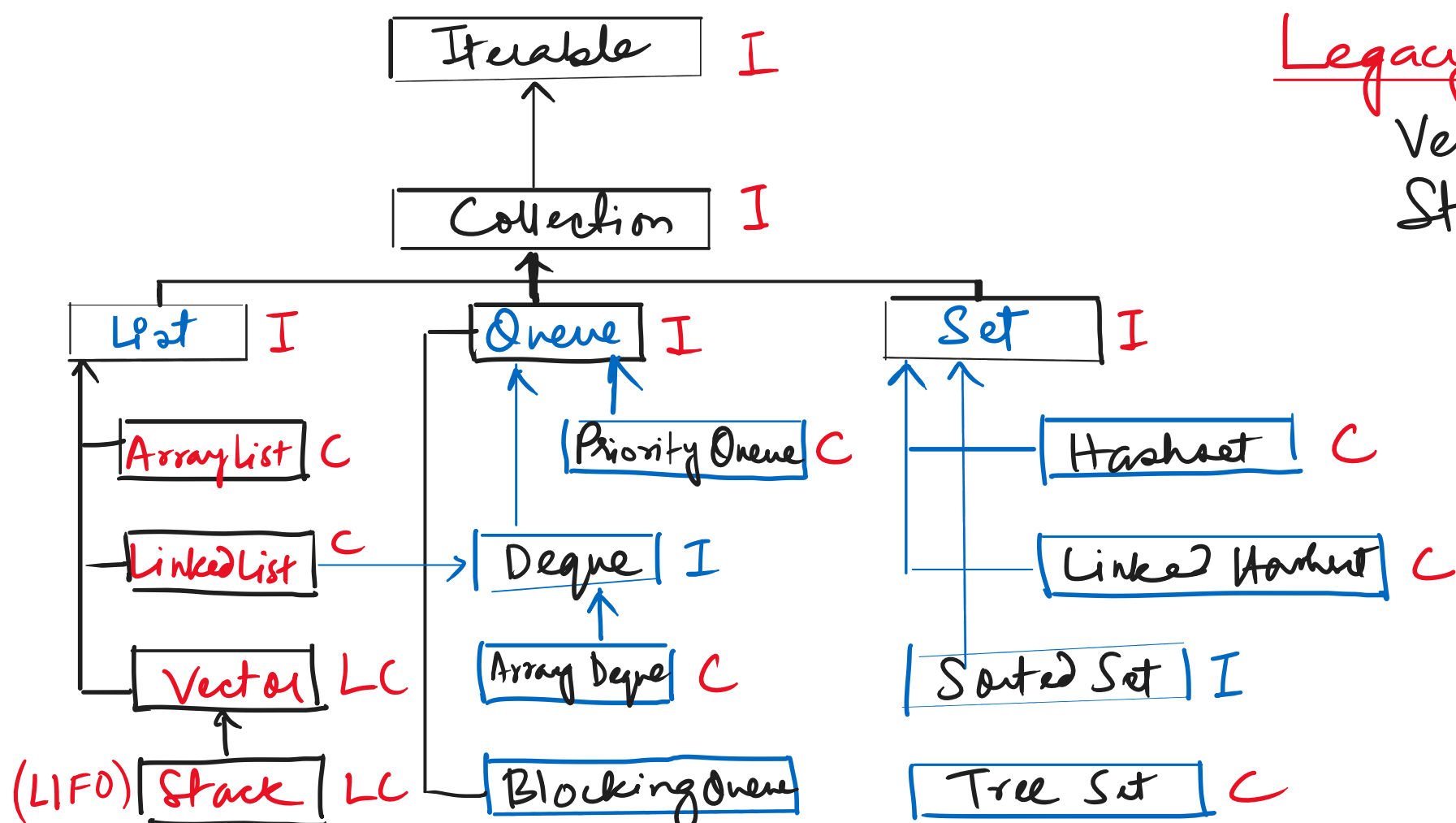
Collections Framework (Java.util)

* It is Java's standard toolkit to store, manage and process groups of objects dynamically.

→ [Collection] Interface Collections [Class]

* Everything else (List, Set, Queue) is built on top of it

***** Map doesn't belong to this Collection Interface.
 [Separate Concept] <K, V> pairs



Legacy Classes ⇒
 Vector {
 Stack {
 } Java 1.0